

## SESSION 3

# RAD-447 Radionuclide Imaging

## PET, Attenuation Correction & Hybrid Imaging

From positron annihilation and coincidence detection to PET/CT, SPECT/CT and PET/MRI

|                      |                                                 |
|----------------------|-------------------------------------------------|
| Course               | RAD-447 Radionuclide Imaging / Nuclear Medicine |
| Date                 | Sept, 2026                                      |
| Lecturer             | A. A. Zubair                                    |
| Suggested study time | 180–240 minutes                                 |

### Learning Objectives

By the end of this session, the student should be able to:

1. explain the basic principle of positron emission tomography (PET);
2. describe positron annihilation and the production of two 511-keV photons;
3. explain coincidence detection and the concept of a line of response;
4. compare PET with SPECT at a conceptual level;
5. explain attenuation and why attenuation correction is required; and
6. describe the roles of PET/CT, SPECT/CT and PET/MRI in hybrid imaging.

### Key Concept

**PET localizes annihilation events by coincidence rather than by a lead collimator.** A positron-emitting radionuclide produces a positron. After the positron encounters an electron, annihilation produces two photons of 511 keV traveling in nearly opposite directions. Detection of the pair within a short timing interval defines a **line of response** through the patient.

## 1 Session Orientation

Session 1 introduced radionuclides, radiopharmaceuticals and radioactive decay. Session 2 followed single gamma photons through the gamma camera and into planar/SPECT images. Session 3 now introduces a different localization principle: **PET uses pairs of annihilation photons detected in coincidence rather than a lead collimator to determine useful event geometry.**

A useful way to organize the session is:

**positron emission → positron travel → annihilation → paired 511-keV photons → coincidence/LOR → corrections → reconstruction → hybrid localization**

#### Why this sequence matters

PET image quality depends on more than tracer uptake. The detector must identify valid coincidence events, corrections must account for effects such as attenuation, scatter and random coincidences, and the reconstructed PET distribution must be accurately aligned with CT or MRI when hybrid imaging is used. A problem at any one of these stages can change lesion conspicuity or apparent tracer concentration.

## 2 What Is PET?

**Positron Emission Tomography (PET)** is a tomographic nuclear-medicine technique that uses radionuclides which decay by positron emission. Common positron-emitting radionuclides used in PET include:

- fluorine-18 (F-18);
- carbon-11 (C-11);
- nitrogen-13 (N-13); and
- oxygen-15 (O-15).

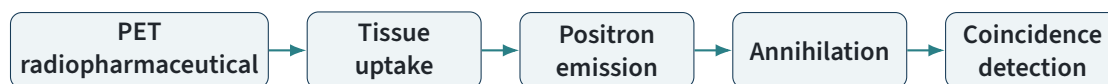
One of the most widely used PET radiopharmaceuticals is **F-18 fluorodeoxyglucose (FDG)**. FDG is a glucose analogue, so its distribution can provide information about tissue glucose metabolism.

#### Clinical / Practical Link

PET is therefore not merely a method of finding “radioactive spots.” The diagnostic meaning comes from the **biological behaviour of the radiopharmaceutical**. With FDG, the image reflects patterns of glucose use rather than anatomy alone.

### 2.1 From tracer uptake to a PET image

The overall process can be summarized as:



The scanner accumulates a very large number of coincidence events from many directions around the patient. Reconstruction algorithms then estimate the three-dimensional distribution of tracer activity.

Modern PET detector modules typically use dense scintillation crystals coupled to photodetectors. The crystal converts the 511-keV photon interaction into light, while the photodetector converts that light into an electrical signal. The system records the interaction position, energy and timing information needed for coincidence processing. The exact detector materials and electronics vary between scanner designs, but the imaging logic remains the same.

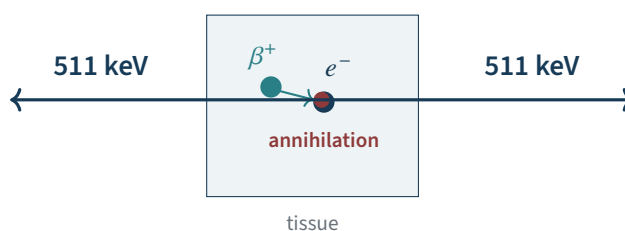
### 3 Positron Emission and Annihilation

A **positron** is the antimatter counterpart of the electron. After emission from the nucleus, it travels a short distance through tissue, losing kinetic energy through interactions. Eventually it encounters an electron.

When the positron and electron interact, **annihilation** occurs. Their mass is converted into photon energy.

The key physical outcome is:

- two photons are produced;
- the photons travel approximately  $180^\circ$  apart; and
- each photon has an energy of **511 keV**.



#### Exam focus

The PET scanner does **not** detect the emitted positron directly as the imaging signal. The clinically useful external signal is the **pair of 511-keV annihilation photons** produced after the positron meets an electron.

#### Positron range and spatial resolution

The positron is emitted at one location but annihilates only after traveling a short distance through tissue. Therefore, the annihilation point is not exactly identical to the site of nuclear decay. The average travel distance depends on positron energy and the surrounding tissue. This **positron range** is one physical limit on PET spatial resolution. A second fundamental limit is **non-collinearity**: the two annihilation photons are emitted nearly, but not perfectly,  $180^\circ$  apart. Both effects blur the relationship between the original nuclear decay site and the reconstructed event location.

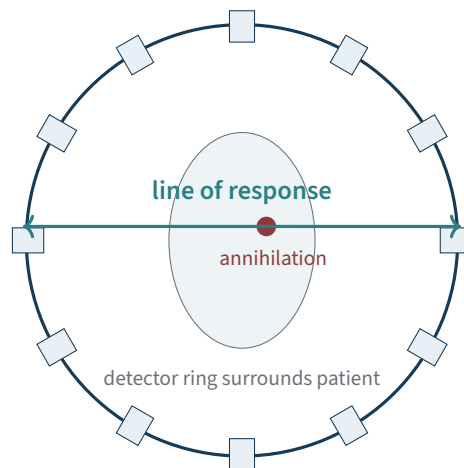
### 4 PET Detector Ring and Coincidence Detection

Unlike the conventional gamma camera described in Session 2, a PET scanner uses detector elements arranged around the patient. When two detectors register 511-keV photons within a sufficiently short timing interval, the system can treat the detections as a candidate pair from the same annihilation event.

This process is called **coincidence detection**.

#### 4.1 The line of response

If a valid pair is detected, the annihilation event is assumed to have occurred somewhere along the line connecting the two detectors. This line is called the **line of response (LOR)**.



One coincidence event defines only a line, not an exact point. By collecting many LORs from many annihilation events and many directions, the scanner can reconstruct the tracer distribution.

#### Electronic versus physical collimation

SPECT uses a physical collimator to accept photons traveling in useful directions. PET instead uses the geometric relationship between a detected photon pair to define an LOR. This is sometimes described conceptually as **electronic collimation**.

Because PET does not require the same type of mechanical lead collimator used in conventional SPECT, a much larger fraction of potentially useful emitted photons can contribute to localization. This is one reason PET can achieve high detection sensitivity, although practical performance still depends on detector efficiency, geometry, count rate, corrections and reconstruction.

## 4.2 Coincidence timing

The detector system uses a short timing interval to decide whether two detections are sufficiently close in time to be treated as a pair. This interval is called the **coincidence timing window**.

Coincidence processing therefore depends on both **energy** and **timing**. Events with inappropriate measured energy or implausible timing can be excluded or corrected statistically. This becomes increasingly important at high count rates, when unrelated detections are more likely to occur close together in time.

#### True, scatter and random coincidences

Not every accepted pair represents ideal geometry. A **true coincidence** is the desired pair from one annihilation event in which neither photon has undergone clinically significant scatter before detection. A photon may undergo Compton scatter and change direction before detection, creating a **scattered coincidence**; the resulting LOR no longer passes accurately through the annihilation site. Two photons from unrelated annihilations may also fall within the coincidence timing window and be paired accidentally, producing a **random coincidence**. Modern PET systems estimate and correct for these unwanted events during data processing and reconstruction.

## 5 SPECT and PET Compared

Both SPECT and PET produce tomographic nuclear-medicine images, but their detector physics differ substantially.

| Feature                    | SPECT                             | PET                                      |
|----------------------------|-----------------------------------|------------------------------------------|
| Radiation detected         | Single gamma photons              | Paired annihilation photons              |
| Detection concept          | Gamma camera with collimator      | Detector ring with coincidence detection |
| Representative energy      | About 140 keV for Tc-99m          | 511 keV per annihilation photon          |
| Common radionuclide        | Tc-99m                            | F-18                                     |
| Main functional emphasis   | Perfusion / organ function        | Metabolism / molecular activity          |
| Typical source examples    | Bone, heart, brain, renal studies | Oncology, brain and cardiac PET          |
| Tomographic reconstruction | Yes                               | Yes                                      |

#### Common Error

Do not say that PET is simply “a more advanced gamma camera.” PET and SPECT are both emission-tomography techniques, but they use different radionuclide decay processes and different localization principles.

## 6 Attenuation: Why Detected Counts Are Not Uniform

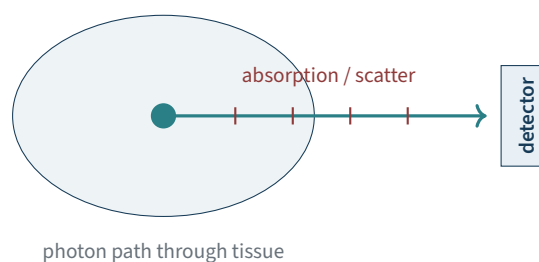
As photons travel through the body, some are absorbed or scattered before they reach the detector. This loss or redirection of photons is called **attenuation**.

Attenuation can produce:

- reduced detected counts;
- apparent areas of decreased activity;
- artifacts; and
- reduced quantitative accuracy.

The amount of attenuation depends on the material the photons traverse and the path length through that material. Consequently, activity deep within the body may be represented differently from identical activity close to the surface if no correction is applied.

In PET, both 511-keV photons must survive the paths from the annihilation site to their respective detectors for an unscattered true coincidence to be recorded. Longer paths through dense tissue therefore have a lower probability of successful detection. Without attenuation correction, the reconstructed activity distribution would be systematically biased by patient size, body composition and event location.



## 7 Attenuation Correction

**Attenuation correction** attempts to compensate for the unequal loss of photons along different paths. Broad approaches to attenuation correction include:

- CT data;
- transmission scanning; and
- mathematical models.

Modern hybrid systems commonly use CT information to estimate tissue attenuation. This is one reason CT contributes to PET/CT and SPECT/CT even when the primary clinical question is functional rather than structural. For PET/CT, CT Hounsfield values are converted into an attenuation map appropriate for the 511-keV PET photons. The PET reconstruction then uses this map to compensate for the different photon losses expected along different LORs. Because the CT and PET datasets may be acquired at different moments, **misregistration** from breathing, patient movement or changing organ position can cause incorrect attenuation correction in a localized region.

### Why attenuation correction matters

Attenuation correction is not merely a cosmetic image-processing step. It can affect lesion conspicuity, artifact appearance and quantitative accuracy. Incorrect attenuation information can therefore create misleading corrected images.

### Common Error

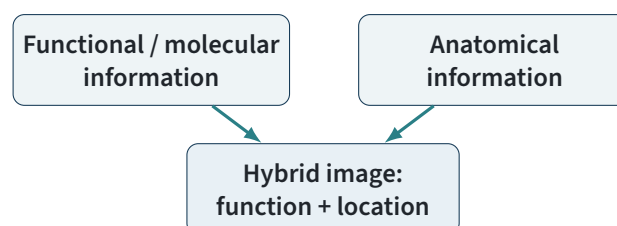
Do not assume that attenuation-corrected PET is automatically more accurate everywhere. If the attenuation map is wrong or misregistered with the emission data, the correction itself can create an apparent hot or cold abnormality. Always consider the non-attenuation-corrected data and the CT/PET alignment when a suspicious correction artifact is possible.

### Practical example

Consider two regions containing the same tracer concentration. One lies near the body surface and one lies deep within the abdomen. The deeper photons traverse more tissue before reaching the detectors and are more likely to be attenuated. Without correction, the deeper region may appear falsely less active.

## 8 Hybrid Imaging

Hybrid imaging combines functional or molecular information with anatomical information. The central idea is simple:



Hybrid imaging can improve anatomical localization of abnormal uptake and can provide data used for attenuation correction.

## 8.1 SPECT/CT

**SPECT/CT** combines:

- SPECT physiological / functional information; and
- CT anatomical information.

A focus of uptake that is difficult to localize on SPECT alone can often be assigned more confidently to a specific bone, joint, organ or soft-tissue structure when registered with CT.

## 8.2 PET/CT

**PET/CT** combines:

- PET metabolic / molecular information; and
- CT anatomical information.

Important benefits include:

- improved lesion localization;
- greater diagnostic confidence;
- attenuation correction; and
- assistance with staging.

### Clinical / Practical Link

A focus of increased FDG uptake may indicate biologically active tissue, but PET alone may not show the anatomical boundary clearly. CT can identify whether that focus lies in a lymph node, lung, bone, bowel or another structure. The value of PET/CT therefore comes from the **relationship between the two data sets**, not from viewing them as unrelated scans.

## 8.3 PET/MRI

**PET/MRI** combines PET molecular information with MRI's strong soft-tissue contrast. Potential applications include brain tumours, neurological disorders, prostate imaging, selected whole-body oncology applications and paediatric imaging.

Attenuation correction is more technically challenging in PET/MRI than in PET/CT because MRI signal intensity is not a direct measure of X-ray attenuation or 511-keV photon attenuation. PET/MRI systems therefore use MR-based segmentation, atlas/model information and other dedicated methods to estimate attenuation.

### Time-of-flight PET

Some modern PET systems use very precise timing differences between the two detected annihilation photons. If one photon arrives slightly earlier than the other, the system can estimate where along the LOR the annihilation was more likely to have occurred. This is called **time-of-flight (TOF) PET**. The approximate localization uncertainty along the LOR is related to timing resolution by

$$\Delta x \approx \frac{c \Delta t}{2},$$

where  $c$  is the speed of light and  $\Delta t$  is the coincidence timing uncertainty. TOF does not replace coincidence

detection; it adds probabilistic position information along the LOR and can improve noise properties and reconstruction convergence, particularly in larger patients.

## 9 PET and Hybrid Imaging in Practice

For the radiographer or nuclear-medicine technologist, hybrid PET studies require careful coordination of patient preparation, tracer administration, uptake conditions, positioning and multimodality acquisition. With FDG, factors such as recent food intake, blood glucose level, strenuous muscle activity, anxiety-related muscle uptake and the interval between injection and imaging can alter biodistribution and image interpretation. Local protocols therefore standardize these variables as far as possible.

During acquisition, patient movement can degrade PET reconstruction and can also misalign the PET and CT/MR datasets. Metallic objects, dense contrast material, respiratory motion and differences in arm position can influence attenuation correction or registration. These issues make patient preparation and positioning part of quantitative image quality rather than merely matters of comfort.

### Practical hybrid-imaging principle

When reviewing a PET/CT study, think of three linked datasets: the **emission PET data**, the **anatomical CT data**, and the **attenuation-corrected PET reconstruction**. A finding should make sense across these datasets. Unexpected focal uptake or photopenia that corresponds to PET/CT misalignment or unusual CT material should raise the possibility of an attenuation-correction artifact.

## 10 What the Student Should Connect Across Sessions

Session 1 established the radionuclide and radiopharmaceutical principles. Session 2 explained single-photon gamma-camera imaging and SPECT. Session 3 completes the detector-physics comparison by showing how PET uses annihilation photon pairs and coincidence geometry.

|                       | Session 1                                    | Session 2                                                     | Session 3                                                              |
|-----------------------|----------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|
| Central question      | What is the tracer and why does it localize? | How are single gamma photons turned into planar/SPECT images? | How are positron annihilation events localized and fused with anatomy? |
| Representative tracer | Tc-99m compounds / FDG introduced            | Tc-99m imaging emphasized                                     | F-18 FDG emphasized                                                    |
| Detector principle    | External detection introduced                | Collimator + gamma camera                                     | Coincidence detector ring                                              |
| Output                | Functional distribution                      | Planar and SPECT images                                       | PET and hybrid images                                                  |

### Key Takeaways

- PET uses positron-emitting radionuclides; F-18 FDG is a major PET radiopharmaceutical used to image patterns of glucose metabolism.
- Positron-electron annihilation produces two photons of 511 keV traveling in nearly opposite directions.
- Coincidence detection defines a line of response between two detectors.
- PET and SPECT are both tomographic emission techniques, but their event-localization physics differ.

- Attenuation reduces detected counts and may produce artifacts or quantitative error.
- CT data can be used for attenuation correction and anatomical localization in hybrid systems.
- SPECT/CT combines physiological information with CT anatomy; PET/CT combines metabolic or molecular information with CT anatomy.
- PET/MRI combines PET molecular information with the excellent soft-tissue contrast of MRI.
- True, scattered and random coincidences have different effects on PET data and must be distinguished conceptually.
- TOF PET uses small arrival-time differences to add approximate localization information along an LOR.

## Key Terms

### **PET**

Positron Emission Tomography; tomographic imaging based on positron-emitting radionuclides and coincidence detection.

### **FDG**

Fluorodeoxyglucose; an F-18-labelled glucose analogue used to image patterns of glucose metabolism.

### **Annihilation**

Interaction of a positron with an electron resulting in conversion of their mass into photon energy.

### **511-keV photon**

Energy of each of the two photons produced in the standard PET annihilation event described in the lecture.

### **Coincidence detection**

Detection of two annihilation photons within a short timing window.

### **Line of response**

Line connecting the paired PET detectors for an accepted coincidence event.

### **Attenuation**

Loss or redirection of photons as they pass through tissue.

### **Attenuation correction**

Compensation for unequal photon loss along different paths through the body.

### **Hybrid imaging**

Registration or integrated acquisition of functional/molecular and anatomical imaging information.

### **True coincidence**

Desired detection of the two photons from one annihilation event without misleading scatter geometry.

### **Random coincidence**

Accidental pairing of photons from unrelated annihilation events within the timing window.

### **TOF PET**

PET that uses coincidence timing differences to estimate the likely annihilation position along the LOR more precisely.

## Self-Assessment Questions

1. Define PET and name four positron-emitting radionuclides used in PET.
2. Explain why F-18 FDG can provide information about tissue metabolism.
3. Describe the sequence from positron emission to the production of two 511-keV photons.
4. What is coincidence detection, and what is a line of response?
5. Compare the event-localization principles of SPECT and PET.
6. Define attenuation and state four consequences associated with attenuation.
7. Explain how CT contributes to attenuation correction in PET/CT or SPECT/CT.
8. State four advantages of PET/CT associated with PET/CT.
9. Explain the complementary roles of PET and MRI in PET/MRI.
10. Why can incorrect attenuation information potentially produce misleading corrected images?
11. Distinguish true, scattered and random coincidences.
12. Explain the principle of TOF PET and what additional information it contributes to a line of response.

## Series Navigation

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| 1 | Foundations of Radionuclide Imaging                                    |
| 2 | Gamma Camera, Planar Imaging & SPECT                                   |
| 3 | <b>PET, Attenuation Correction &amp; Hybrid Imaging</b> — this session |
| 4 | Clinical Radionuclide Imaging Applications                             |
| 5 | Image Quality, Radiation Protection, QC, Artifacts & Future Directions |

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